CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY (CHARUSAT)

Chandubhai S. Patel Institute of Technology (CSPIT)

Department of Computer Science & Engineering

ITUE203 – Web Development Frameworks

PRACTICAL 1

Initiation & Planning of the StudentHub Portal

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Semester / Academic Year	3 / 2026-27
Date of Submission	
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1. Aim

To initiate the development of the StudentHub portal by analysing project requirements, identifying user roles and modules, designing the sitemap and low-fidelity wireframes, organising the project folder structure, and preparing the README file and GitHub repository for future development.

2. Problem Statement

Initiate the semester-long StudentHub portal by identifying the problem scope, user roles, key modules, navigation flow and a minimum of 10 pages. Create a sitemap, a low-fidelity wireframe, the project folder structure, a README file and a GitHub repository.

3. Requirement Analysis

3.1 Problem Scope

Students of the CSE department need a single place to see their timetable, enrolled subjects, attendance, results, assignments and campus notices, while administrators need a secure way to manage student records, events and user accounts. StudentHub is a web portal built with PHP, MySQL, HTML, CSS and JavaScript that provides these modules with role-based access.

3.2 Functional Requirements

- **Public portal:** home page with announcements, image slider, FAQ and guidelines; contact/inquiry page.
- **Accounts:** student self-registration and login with hashed passwords; logout.
- **Role-based access:** students are taken to the student dashboard, administrators to the Admin Console; other users opening the Admin Console see an access-denied (403) page.
- **Student modules:** dashboard, profile with password change, courses, attendance, assignments, results, achievements, notices, directory.
- **Administration:** manage student records (add, search, filter, delete, CSV export), publish events with posters, view user accounts and the security audit log.

3.3 Non-Functional Requirements

- **Usability:** consistent navigation and layout designed for desktop and mobile screens, with light and dark themes.
- **Accessibility:** semantic HTML structure, labelled form fields and visible keyboard focus.
- **Security:** bcrypt password hashing, prepared statements for queries that use input, server-side role checks and escaped output.
- **Maintainability:** shared stylesheet and a reusable sidebar include across pages.

4. User Roles

Role	Responsibilities	Status
Visitor	View the home page, register, log in, send an inquiry.	Implemented
Student	View dashboard, timetable, attendance, results, assignments, events; manage profile and password.	Implemented
Faculty	Manage courses, upload assignments, update attendance, check results.	Planned – not implemented in the main application
Admin	Manage student records, events and user accounts; export CSV; review the audit log.	Implemented

5. Key Modules

Module	Pages	Purpose
Public portal	index.php, contact.php	Landing page and department contact
Authentication	login.php, register.php, logout.php	Accounts, sessions and role-based redirect
Student dashboard	dashboard.php	Summary cards, timetable, upcoming events
Profile	profile.php	Student details and password change
Academics	courses.php, attendance.php, assignments.php, results.php, achievements.php	Subjects, attendance, assignment status, results, achievements
Campus	notifications.php, directory.php	Notices and student directory
Administration	admin_dashboard.php (5 tabs)	Students, events, users and audit log
Access control	403.php	Shown when a non-admin opens the Admin Console

6. Page Inventory

The portal has 15 screens plus a logout action, which satisfies the requirement of at least 10 pages.

#	Page	File	Access
1	Home Portal	index.php	Public
2	Login	login.php	Public
3	Register	register.php	Public
4	Contact Us	contact.php	Public
5	Student Dashboard	dashboard.php	Student

#	Page	File	Access
6	My Profile	profile.php	Student
7	Courses	courses.php	Student
8	Attendance	attendance.php	Student
9	Assignments	assignments.php	Student
10	Results	results.php	Student
11	Achievements	achievements.php	Student
12	Notices	notifications.php	Student
13	Directory	directory.php	Student
14	Admin Console	admin_dashboard.php	Admin
15	Access Denied	403.php	Non-admin users
16	Logout (action)	logout.php	Logged-in users

7. Sitemap & Navigation Flow

Figure 1 shows the sitemap of StudentHub. It was drawn from the pages and links that exist in the application.

- A visitor starts at the Home Portal and can open Login, Register or Contact Us.
- Login verifies the password and checks the role: students go to the Student Dashboard, administrators to the Admin Console.
- After registering, a new student is signed in automatically and taken to the dashboard.
- Inside the student area a sidebar groups the pages into Overview, Academics, Campus and Account.
- The Admin Console has five tabs: Overview, Students, Events, Users and Audit Logs.
- A non-admin who opens the Admin Console is shown the Access Denied (403) page.
- Logout ends the session and returns to the login page.

8. Low-Fidelity Wireframes

Greyscale low-fidelity wireframes were prepared for the key screens. They show layout and content hierarchy only; they are design drawings, not screenshots of the application.

- Figure 2 – Student dashboard (desktop): sidebar, welcome header, summary cards, weekly timetable and upcoming events.
- Figure 3 – Home page and login (desktop): public landing page and the split login layout.
- Figure 4 – Mobile: dashboard, open navigation drawer and login on a phone-sized screen.

Figure 1 – Sitemap & Navigation Flow

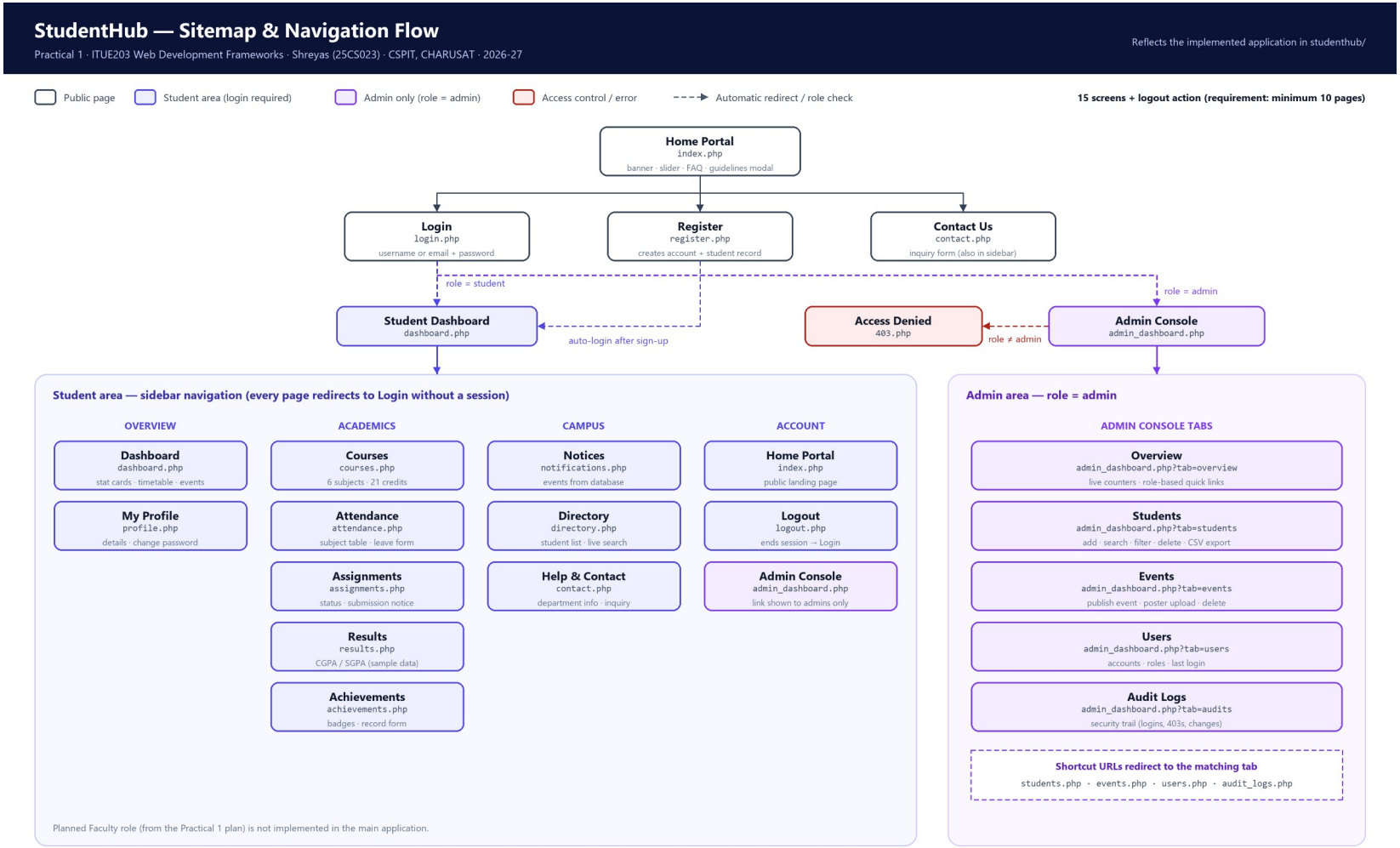


Figure 1: Sitemap and navigation flow of StudentHub

Figure 2 – Wireframe: Student Dashboard (desktop)

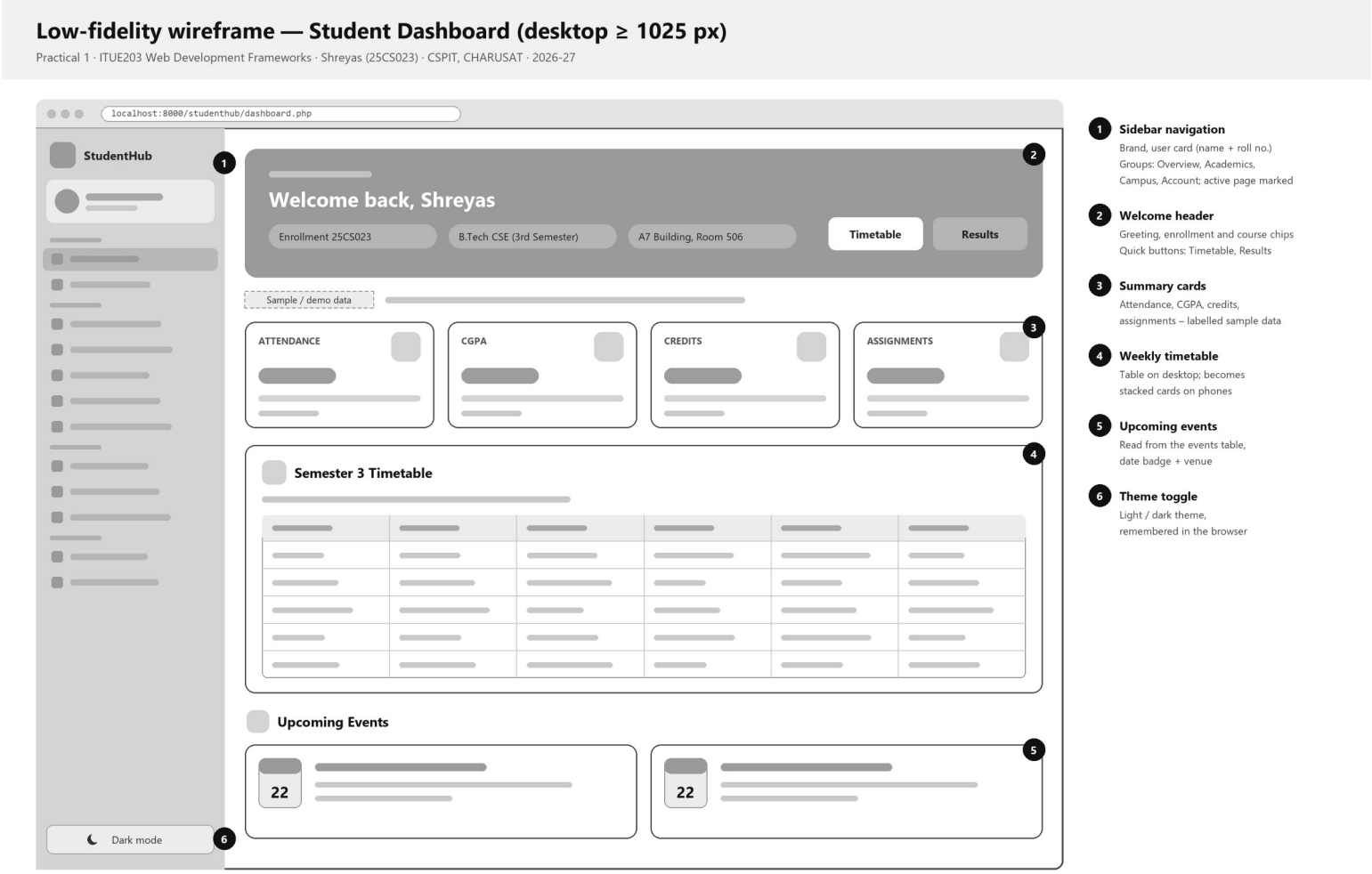


Figure 2: Low-fidelity wireframe of the student dashboard (desktop)

Figure 3 – Wireframes: Home Page and Login (desktop)

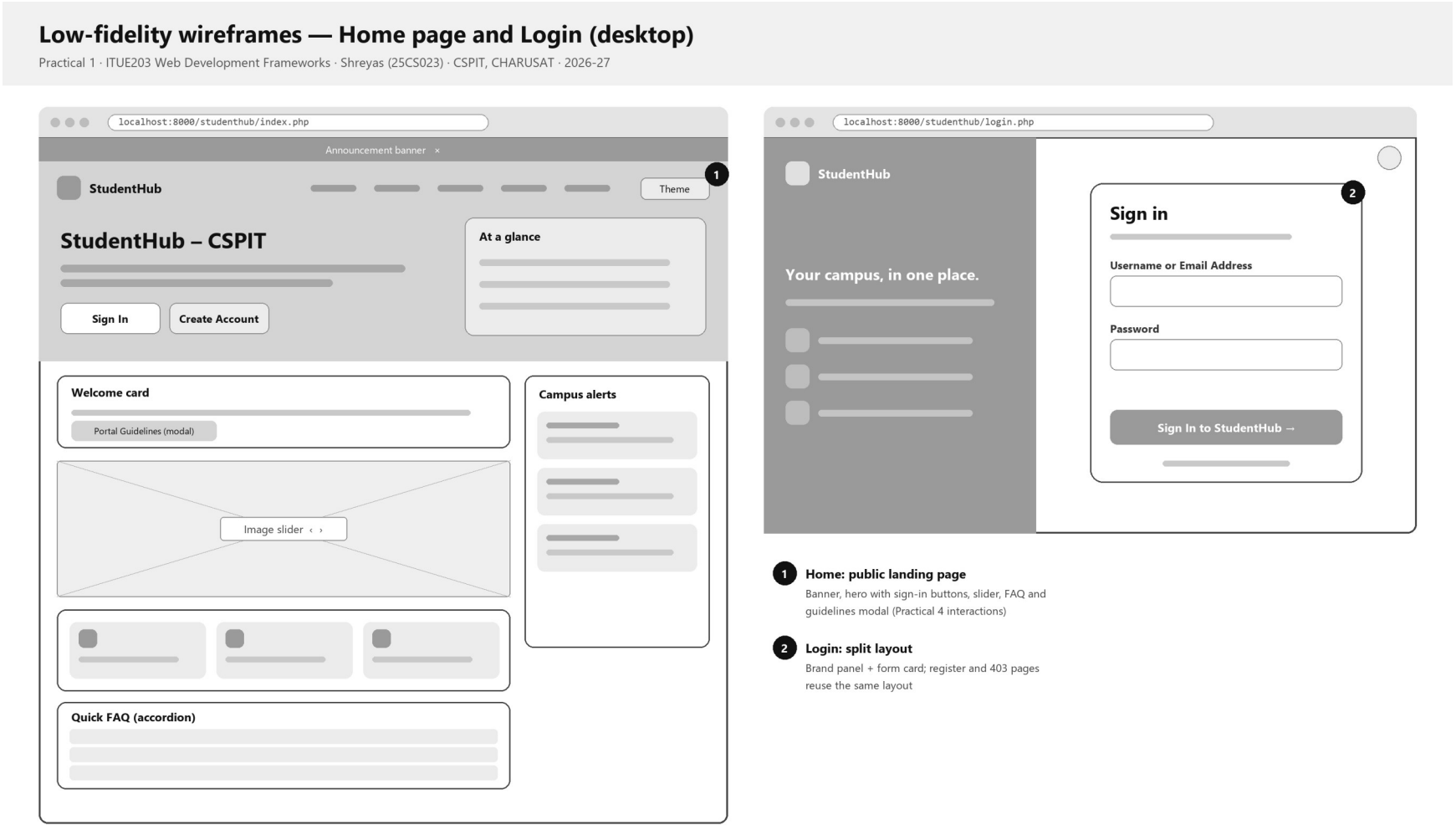


Figure 3: Low-fidelity wireframes of the home page and login (desktop)

Figure 4 – Wireframes: Mobile Screens

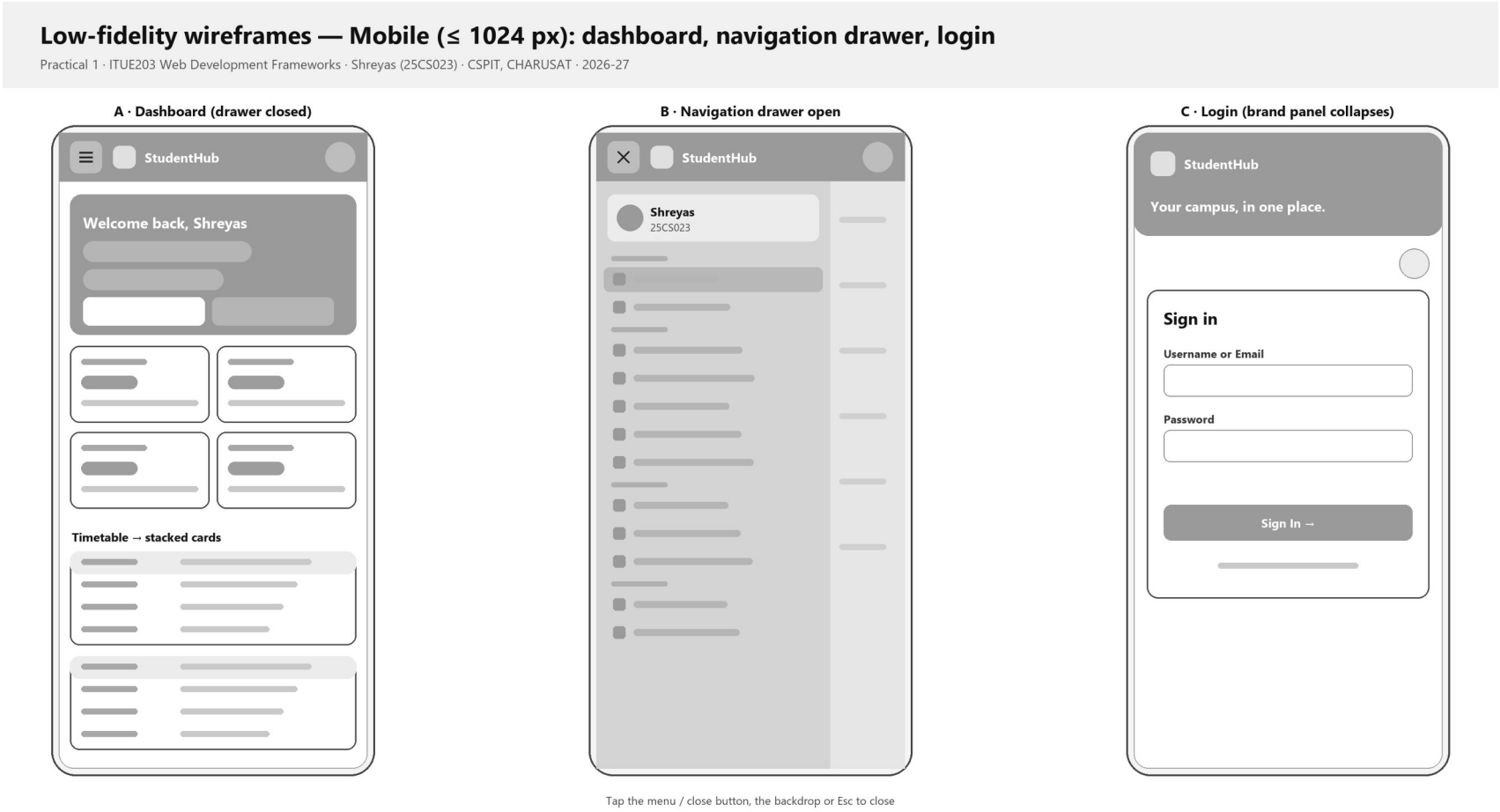


Figure 4: Low-fidelity wireframes of the mobile dashboard, navigation drawer and login

9. Project Folder Structure

```
StudentHub/
├── studenthub/          Main application (PHP + MySQL)
├── Practical-1/         Planning: sitemap, wireframes, README
├── Practical-2 … 15/    Practical-wise deliverables
├── database/            Safe demo SQL for the main database
├── tools/               Command-line helper to create demo accounts
├── README.md
└── TEST_REPORT.md

Practical-1/
├── README.md
├── sitemap.svg / sitemap.png
├── wireframe-1-dashboard-desktop.svg / .png
├── wireframe-2-home-login-desktop.svg / .png
├── wireframe-3-mobile.svg / .png
├── PRAC_1_Sitemap_25CS023.pdf       (sitemap, 1 page)
├── PRAC_1_Wireframes_25CS023.pdf    (wireframes, 3 pages)
├── PRAC_1_ITUE203_25CS023.docx      (this report)
├── PRAC_1_ITUE203_25CS023.pdf       (PDF copy of this report)
├── screenshots/                     (GitHub screenshot – to be added)
└── source/build_diagrams.py, export_pdfs.py
```

10. README File

The repository contains a root README.md that describes the project, its features, local setup and the status of each practical. Practical-1/README.md records the objective, problem statement, user roles, sitemap, wireframes and folder structure of this practical.

11. Source Code Evidence

The excerpts below are copied unchanged from the StudentHub application in the repository (studenthub/ folder; Practical-15 holds an identical copy). Line numbers match the files; only the common leading indentation is removed. They show how the planned page structure and navigation flow are implemented. Practical 1 is a planning practical: source code is not one of its required deliverables, so this section is supporting evidence only.

File	Lines	Shows
studenthub/index.php	18–27	Document structure and metadata of the home page
studenthub/index.php	36–65	Header, brand, theme toggle, hamburger button and main navigation
studenthub/sidebar.php	8–32	Navigation map of the signed-in area and the skip link
studenthub/sidebar.php	67–78	Loop that renders the menu and the admin-only link

11.1 Home page: document structure and metadata

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>StudentHub - CSPIT Charusat Portal</title>
    <link rel="stylesheet" href="assets/css/home.css?v=1">
    <script src="assets/js/theme.js"></script>
</head>
<body>
```

- **Document type and language:** <!DOCTYPE html> selects standards mode; lang="en" tells browsers and screen readers the page language.
- **Metadata:** UTF-8 character set, a viewport tag for correct scaling on phones, and a descriptive page title.
- **Stylesheet and script:** home.css styles the public page (it imports the shared design tokens, web font and Bootstrap Icons); theme.js applies the saved light/dark theme before the page is drawn.

11.2 Home page: header and navigation

```
<header class="site-header">
    <div class="topnav">
        <a class="site-brand" href="index.php">
            <span class="brand-mark" aria-hidden="true"><i class="bi bi-mortarboard-fill"></i></span>
            <span><strong>StudentHub</strong><small>CSPIT · CHARUSAT</small></span>
        </a>
        <button class="theme-toggle" id="theme-toggle-btn" type="button"><i class="bi bi-moon-stars" aria-hidden="true"></i> Dark</button>
        <button class="hamburger" id="hamburger-menu" type="button" aria-label="Toggle navigation menu"><i class="bi bi-list" aria-hidden="true"></i></button>

        <!-- Navigation (collapses behind the hamburger on phones – Practical 4) -->
        <nav class="hero-nav" aria-label="Main navigation">
            <ul id="nav-links">
                <li><a href="index.php" class="active" aria-current="page">Home</a></li>
                <?php if ($is_logged_in): ?>
                    <li><a href="dashboard.php">Dashboard</a></li>
                    <li><a href="profile.php">My Profile</a></li>
                    <li><a href="courses.php">Courses</a></li>
                    <li><a href="attendance.php">Attendance</a></li>
                    <?php if ($user_role === "admin"): ?>
                        <li><a href="admin_dashboard.php" class="nav-admin"><i class="bi bi-shield-lock" aria-hidden="true"></i> Admin Console</a></li>
                    <?php endif; ?>
                    <li><a href="logout.php">Logout</a></li>
                <?php else: ?>
                    <li><a href="login.php">Login</a></li>
                    <li><a href="register.php">Register</a></li>
                    <li><a href="dashboard.php">Dashboard</a></li>
                    <li><a href="contact.php">Contact Us</a></li>
                <?php endif; ?>
            </ul>
        </nav>
```

- **Semantic landmarks:** <header> holds the site brand and a <nav> labelled "Main navigation" for assistive technology; icons are hidden from screen readers with aria-hidden.
- **Navigation flow:** visitors see Home, Login, Register, Dashboard and Contact Us; signed-in users see Dashboard, My Profile, Courses, Attendance and Logout, and admins also get the Admin Console link. The PHP session decides which list is printed.
- **Responsive menu:** on narrow screens the list collapses behind the hamburger button (#hamburger-menu), which script.js toggles.

11.3 Signed-in area: navigation map and skip link

```
// Grouped navigation: [file, icon, label, pages that mark it active]
$nav_groups = [
    "Overview" => [
        ["dashboard.php",     "bi-grid-1x2",         "Dashboard",     ["dashboard.php",
"student_dashboard.php"]],
        ["profile.php",       "bi-person-circle",    "My Profile",    ["profile.php"]],
    ],
    "Academics" => [
        ["courses.php",       "bi-journal-bookmark", "Courses",       ["courses.php"]],
        ["attendance.php",    "bi-calendar-check",   "Attendance",    ["attendance.php"]],
        ["assignments.php",   "bi-pencil-square",    "Assignments",   ["assignments.php"]],
        ["results.php",       "bi-bar-chart-line",   "Results",       ["results.php"]],
        ["achievements.php",  "bi-award",            "Achievements",  ["achievements.php"]],
    ],
    "Campus" => [
        ["notifications.php", "bi-megaphone",        "Notices",       ["notifications.php"]],
        ["directory.php",     "bi-people",           "Directory",     ["directory.php"]],
        ["contact.php",       "bi-life-preserver",   "Help & Contact", ["contact.php"]],
    ],
    "Account" => [
        ["index.php",         "bi-house",            "Home Portal",   ["index.php"]],
        ["logout.php",        "bi-box-arrow-right",  "Logout",        []],
    ],
];
?>
<a class="skip-link" href="#main-content">Skip to main content</a>
```

```
<?php foreach ($nav_groups as $group => $links): ?>
    <p class="nav-group-label"><?= $group ?></p>
    <ul>
        <?php foreach ($links as [$href, $icon, $label, $active_pages]): ?>
            <?php $is_active = in_array($current_page, $active_pages, true); ?>
            <li><a href="<?= $href ?>"<?= $is_active ? ' class="active" aria-current="page"' : '' ?>><i
class="bi <?= $icon ?>" aria-hidden="true"></i><span><?= $label ?></span></a></li>
        <?php endforeach; ?>
        <?php if ($group === "Account" && $is_admin): ?>
            <li><a href="admin_dashboard.php" class="nav-admin"><i class="bi bi-shield-lock" aria-
hidden="true"></i><span>Admin Console</span></a></li>
        <?php endif; ?>
    </ul>
<?php endforeach; ?>
```

- **Single navigation map:** $nav_groups lists every signed-in page under Overview, Academics, Campus and Account – the same grouping as the sitemap in Figure 1.
- **Rendering:** the loop prints each group, marks the current page with class="active" and aria-current="page", and adds the Admin Console link only when the session role is admin.
- **Accessibility:** the skip link jumps to <main id="main-content"> on every signed-in page. The public home page (index.php) does not include a skip link.
- **Reuse:** each signed-in page includes sidebar.php, so the menu is defined once.

11.4 Screenshots from VS Code (optional)

OPTIONAL – VS Code screenshot: studenthub/index.php, lines 18–65

Not required for Practical 1. Insert a genuine screenshot of the file open in VS Code, or delete this box.

OPTIONAL – VS Code screenshot: studenthub/sidebar.php, lines 8–32

Not required for Practical 1. Insert a genuine screenshot of the file open in VS Code, or delete this box.

12. GitHub Repository

The StudentHub source code, the practical folders (Practical-1 to Practical-15) and their documentation are kept in the student's own GitHub repository.

Repository URL: https://github.com/TheEmperor08/StudentHub-ITUE203-25CS023

PLACEHOLDER – GitHub repository screenshot (genuine screenshot required, not yet inserted)

Insert a screenshot of the repository home page https://github.com/TheEmperor08/StudentHub-ITUE203-25CS023 showing the repository name, file list and README. A screenshot of a single folder is not sufficient. Save the same image as Practical-1/screenshots/github-repository.png.

13. Tools & Technologies

Area	Used in StudentHub
Front end	HTML5, CSS3 (Grid, Flexbox), JavaScript, Bootstrap 5 (admin console), Bootstrap Icons
Back end	PHP 8.2 (8.2.12, bundled with XAMPP)
Database	MariaDB 10.4 (MySQL-compatible, bundled with XAMPP)
Local server	XAMPP; PHP built-in server (php -S) for local testing
Diagrams	Python scripts in Practical-1/source/: build_diagrams.py (SVG and PNG), export_pdfs.py (PDF)
Version control	Git and GitHub (TheEmperor08/StudentHub-ITUE203-25CS023)

14. Submission Deliverables

The deliverables required for Practical 1 and where each one is provided:

#	Deliverable	Provided as
1	Complete StudentHub project folder	The whole repository: studenthub/ (main application), Practical-1 to Practical-15, database/, tools/, README.md
2	GitHub repository screenshot	Pending – placeholder in Section 12 Genuine screenshot to be inserted and saved as Practical-1/screenshots/github-repository.png
3	GitHub repository link	https://github.com/TheEmperor08/StudentHub-ITUE203-25CS023
4	Sitemap PDF	Practical-1/PRAC_1_Sitemap_25CS023.pdf 1 page, A4 landscape (Figure 1)
5	Low-fidelity wireframe PDF	Practical-1/PRAC_1_Wireframes_25CS023.pdf 3 pages, A4 landscape: student dashboard (desktop), home page and login (desktop), mobile screens (Figures 2–4)

This report is provided as PRAC_1_ITUE203_25CS023.docx, with a PDF copy PRAC_1_ITUE203_25CS023.pdf in the same folder.

15. Conclusion

In this practical the StudentHub portal was planned: the problem scope and requirements were analysed, the user roles and modules were identified, and a page inventory of 15 screens was prepared. A sitemap and low-fidelity wireframes for desktop and mobile were created, and the project folder structure and README were organised. A GitHub repository was set up under the student's own account to hold the project.